Motivation: Market Regimes

- Market regimes describe distinct periods of return behavior, but the underlying states are **unobserved**.
- Identifying these regimes may help characterize changing market conditions and inform investment decisions.
- Common models for modeling financial returns such as Geometric Brownian Motion assumes constant drift and volatility, which limits its ability to represent these changes.
- Therefore, unsupervised learning algorithms that explicitly model regime-switching is a suitable model for this type of problems.

Hidden Markov Model

- Hidden Markov Model (HMM) is an unsupervised learning probabilistic model.
- Assume that the sequences of observations (market data) $x_{1:T}$ are generated by latent states (regimes) $z_{1:T}$, where the latent states evolve according to a first-order Markov chain. [1]
- First-order Markov assumption (i.e. memoryless after 1 time lag)
- Assume observations conditionally independent given latent state

HMM Training

- HMM is learns its parameter with Baum-Welch algorithm, a type of Expectation-Maximization algorithm for HMMs [1].
- **E-step:** Use forward–backward to compute posterior probabilities of hidden states and transitions.
- **M-step:** Update initial probabilities, transition probabilities, and emission parameters using these posterior probabilities as weights.
- Repeat until the log-likelihood converges or the iteration limit is reached.

$$\hat{\theta} = \arg\max_{\theta} \log p_\theta(x_{1:T}), \qquad \theta = (\boldsymbol{\pi}, A, \boldsymbol{\theta}_{\text{emission}}).$$

Baum-Welch algorithm

Algorithm 1 Baum–Welch

Input: Observations $X = (x_1, \ldots, x_T)$, states K, initial parameters $\phi = (\pi, A, \theta)$, tolerance ε, and iteration limit R.

Define $b_i(x_t) = p(x_t \mid z_t = i; \theta_i)$.
Initialize ϕ and compute $\ell_{\text{old}} = \log p(X \mid \phi)$.
for $r = 1, \ldots, R$ **do**
    Initialize $\alpha_1(i) = \pi_i b_i(x_1)$ and $\beta_T(i) = 1$.
    Forward recursion, $t = 2, \ldots, T$: $\alpha_t(j) = b_j(x_t) \sum_{i=1}^{K} \alpha_{t-1}(i) A_{ij}$.
    Backward recursion, $t = T-1, \ldots, 1$: $\beta_t(i) = \sum_{j=1}^{K} A_{ij} b_j(x_{t+1}) \beta_{t+1}(j)$.
    Compute sequence likelihood: $L = \sum_{i=1}^{K} \alpha_T(i)$.
    Compute state posteriors: $\gamma_t(i) = \alpha_t(i)\beta_t(i)/L$.
    Compute transition posteriors, $t = 1, \ldots, T-1$: $\xi_t(i,j) = \alpha_t(i) A_{ij} b_j(x_{t+1}) \beta_{t+1}(j)/L$.
    Update $\pi_i^{\text{new}} = \gamma_1(i)$.
    Update $A_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}$.
    Update emission parameters: $\theta^{\text{new}} = \arg\max_\theta \sum_{t=1}^{T} \sum_{i=1}^{K} \gamma_t(i) \log p(x_t \mid z_t = i; \theta_i)$.
    Set $\phi \leftarrow (\pi^{\text{new}}, A^{\text{new}}, \theta^{\text{new}})$.
    Compute $\ell_{\text{new}} = \log p(X \mid \phi)$.
    if $|\ell_{\text{new}} - \ell_{\text{old}}| \le \varepsilon$ **then**
        break
    end if
    $\ell_{\text{old}} \leftarrow \ell_{\text{new}}$.
end for
return fitted parameters ϕ.

HMM Inference

- Given a sequence of observations and fitted parameters, HMM returns the most probable hidden state sequence via the Viterbi algorithm [1].
- Using dynamic programming to track the best path ending in each state, then backtrack to recover the overall best path.

Viterbi algorithm

Algorithm 2 Viterbi

Input: Observation sequence $X = (x_1, \ldots, x_T)$, initial-state probabilities π, transition matrix A, and emission parameters θ

Initialize $\delta_1(j) = \pi_j p(x_1 \mid z_1 = j)$
for $t = 2, \ldots, T$ **do**
    for $j = 1, \ldots, K$ **do**
        Compute $\delta_t(j) = \max_i \left[\delta_{t-1}(i) A_{ij}\right] p(x_t \mid z_t = j)$
        Store the maximizing previous state: $\psi_t(j) = \arg\max_i \left[\delta_{t-1}(i) A_{ij}\right]$
    end for
end for
Set final state: $z_T^* = \arg\max_j \delta_T(j)$
for $t = T - 1, \ldots, 1$ **do**
    Backtrack: $z_t^* = \psi_{t+1}(z_{t+1}^*)$
end for
return most likely state sequence $z_{1:T}^*$

Graphical Representation of HMM

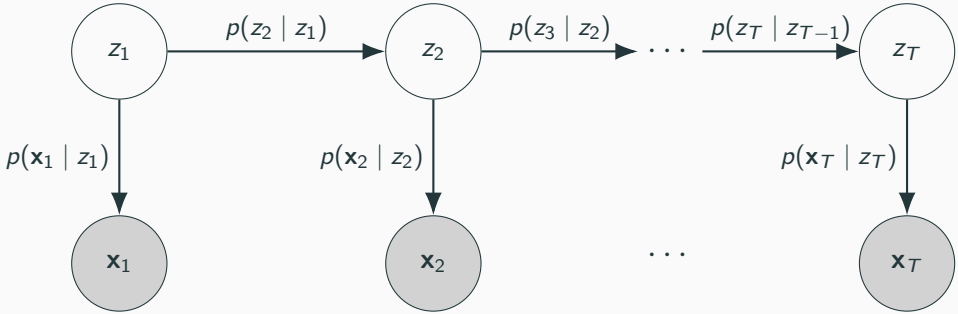


Figure 1: Hidden Markov model as a directed graph.

Emission Models

- The emission distribution $p(x_t|z_t)$ models how each hidden regime gives rise to the observed features.
- Typically, Gaussian distribution is used, parameterized by mean μ and variance σ^2.

Problem:

- Financial asset returns exhibit well-known stylized facts such as heavy tails, skewness, and volatility clustering [2].
- The Gaussian assumption may be restrictive when the distribution of returns within a regime exhibit these characteristics on financial data.

Research Questions

This motivates us to look for more flexibility in the emission model. We consider the following research questions:

- What's the consequence of using Gaussian emission model to represent data that is not Gaussian?
- Does a more flexible emission model improves the detection of regimes? Up until this midterm stage, we consider using Gaussian Mixture Models.
- Does the identified regimes provide useful information for investment decisions?

Recent Research in Emission Representation

- **Heavy-tailed emissions:** Peruilh Bagolini et al. (2026) use multivariate Student-t emissions in a Bayesian HMM to model cryptocurrency returns [3].
- **Flexible parametric distributions:** Foroni et al. (2024) use state-dependent generalized hyperbolic distributions to model financial dependence networks [4].
- **Neural density models:** Lorek et al. (2022) introduce FlowHMM, using normalizing flows to learn each state's emission density [5].

Connection to our study

Flexible emissions can better represent within-state dynamics. We will investigate how more flexible emissions improve regime detection over Gaussian, and the feasibility of neural density models like normalizing flow on financial time series data in the near future.

Gaussian vs. Gaussian mixture emissions

Gaussian HMM

$$p_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$$

- One Gaussian per state.
- Simple mean and variance interpretation.
- Restrictive for skewness and heavy tails.

Gaussian mixture HMM

$$p_k(x) = \sum_{m=1}^{M} w_{km} \mathcal{N}(x; \mu_{km}, \sigma_{km}^2)$$

- Multiple Gaussian components within each state.
- Have greater flexiblility in describing the target distribution.
- More parameters and greater fitting cost.

Challenges in Evaluation

Evaluating regime-switching models on financial data in general remains an open challenge.

Problems:

- We won't be able to evaluate the model on its regime detection capabilities on real financial data because no true regime label exist.
- There is no one standard definition for market regime. Bull/bear markets and high/low volatility describe different aspects of market behavior and can all be considered the definition of different regimes.
- There is generally also no universally accepted standardized benchmarks for regime-switching models in finance. Differences in datasets, regime definitions, and evaluation protocols limit comparisons across studies [6].

These limitations motivate our use of complementary synthetic and real-data experiments.

Evaluation Methodology

Synthetic data: controlled evaluation

- Known regime labels and generating distributions allow direct evaluation of state, emission, and transition recovery.
- Vary emission shape and regime persistence to examine when each model succeeds or fails.

Real data: practical evaluation

- Evaluate performance under market behavior that controlled simulations may not capture.
- Assess out-of-sample distributional fit and whether regime-based signals improve downstream trading performance.

Synthetic experimental design

Generating families

- Gaussian: correctly specified baseline.
- Student-t: heavy tails.
- Trimodal mixture: multimodality.
- Shifted Gamma: skewness.

Common fitting protocol

- 20,000 training; 5,000 test points.
- 10 independent replications.
- 10 starts; retain highest training likelihood.
- At most 1,000 EM updates per start.

- Independent train and test sequences to test models' ability to generalize to another realization
- We do 10 different initializations in an attempt to get a better optima as Baum-Welch is sensitive to initialization [1].

Identical observations for both models; independent train and test sequences. Families are matched in state-specific means and variances within each setting.

Two Types of Synthetic Regimes

We define 2 types of synthetic regimes: First by emissions distribution in each state vary by their mean and hold their standard deviation fixed, and the other where we vary their stardard deviation and hold mean fixed.

	Mean-separated	Volatility-defined
State means	$(-2, 0, 2)$	$(0, 0, 0)$
State standard deviations	$(1, 1, 1)$	$(1, 2, 4)$
Self-transition probability	0.995	0.995

$$A_{kk} = p, \qquad A_{kj} = (1-p)/2 \ (j \neq k), \qquad \pi_k = 1/3.$$

Synthetic Data Generation

1. Sample the initial hidden state from a categorical distribution defined by the initial probabilities,
$$z_1 \sim \text{Categorical}(\boldsymbol{\pi}).$$
2. For each subsequent time step, sample the hidden state according to the transition probabilities of the previous state,
$$z_t \mid z_{t-1} \sim \text{Categorical}(A_{z_{t-1}}).$$
3. The sampled state z_t determines the corresponding emission distribution.
4. Sample the observation from that state-specific distribution,
$$x_t \mid z_t = k \sim p_k(x).$$
5. Repeat for $t = 1, \ldots, T$ to obtain the hidden-state sequence $z_{1:T}$ and observation sequence $x_{1:T}$.

Example of Synthetic Data

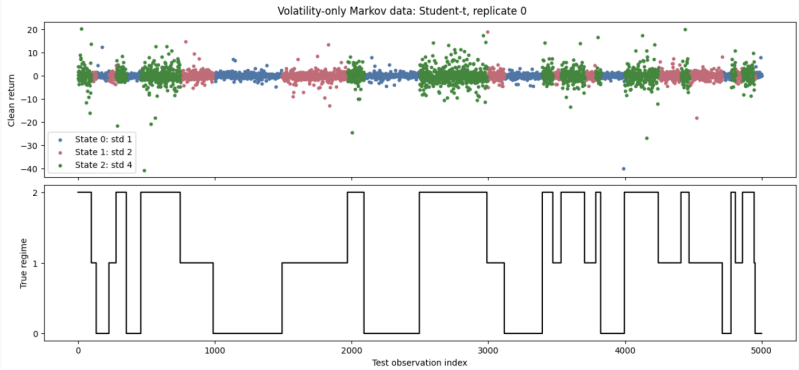


Figure 2: An example of Student-t based emission synthetic data partitioned by regimes

Evaluation Metrics

Measure	What it evaluates
Macro F1 / accuracy	Agreement with known synthetic states
Emission KL divergence	Recovery of each state's generating density
Transition RMSE	Recovery of the generating transition matrix
Decoded switch count	How often the inferred state sequence changes

Problem: Arbitrary Labels

- HMM hidden-state labels are arbitrary.
- The model is not told which state should be called 0, 1, or 2.

True states:	0	1	2
Learned states:	2	0	1

These can represent the same regimes, only with different labels.

State labels must be aligned before evaluation

Solution: Arbitrary Labels

We use a similar approach to Wiedenhoeft et al. by solving a linear assignment problem that maximizes agreement between training Viterbi predictions and true training states [7].

1. Fit the HMM using only training observations.
2. Use known synthetic training labels to find the state mapping which maximizes the total number of matched observations.
3. Freeze this mapping.
4. Apply it to predictions on the test set.

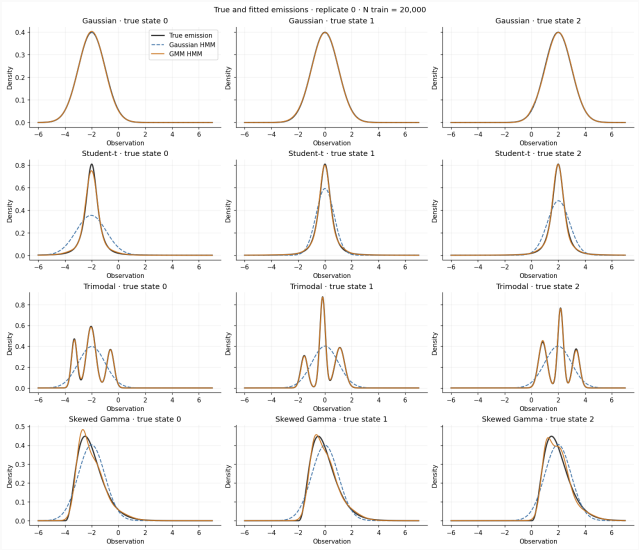
Solved using `scipy.optimize.linear_sum_assignment`.

Mean-Separated Regimes: State and Density Recovery

Family	Model	Macro F1	Accuracy	Emission KL
Gaussian	GHMM	0.9980	0.9980	0.00015
	GMM-HMM	0.9979	0.9980	0.00038
Student-t	GHMM	0.9901	0.9907	0.49280
	GMM-HMM	**0.9991**	**0.9992**	**0.01046**
Trimodal	GHMM	0.9975	0.9976	0.29469
	GMM-HMM	**0.9984**	**0.9985**	**0.00051**
Skewed Gamma	GHMM	0.9971	0.9973	0.08905
	GMM-HMM	**0.9984**	**0.9985**	**0.00679**

- Both models achieve high state-classification performance.
- GMM-HMM substantially improves non-Gaussian density fit.
- Accurate state recovery can coexist with emission misspecification.

Mean-Separated Regimes: Fitted Densities Graph

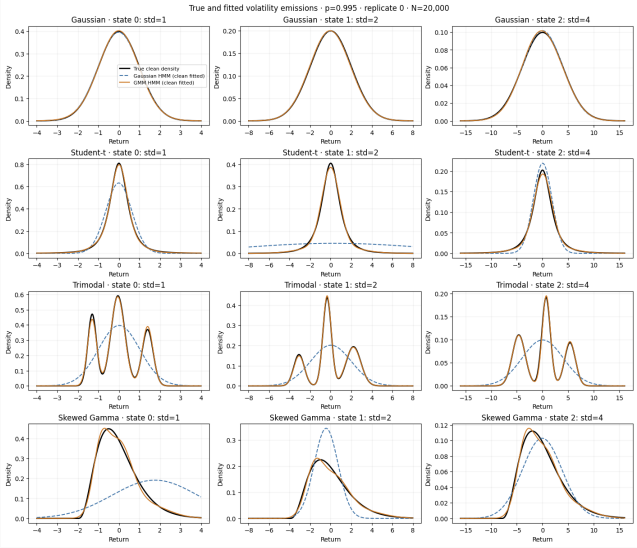


Volatility-Defined Regimes: State and Density Recovery

Family	Model	Macro F1	Accuracy	Emission KL
Gaussian	GHMM	0.9806	0.9813	0.00017
	GMM-HMM	0.9806	0.9812	0.00044
Student-t	GHMM	0.4855	0.5675	1.01907
	GMM-HMM	**0.9527**	**0.9535**	**0.01061**
Trimodal	GHMM	0.7003	0.7484	4.00752
	GMM-HMM	**0.9938**	**0.9941**	**0.00064**
Skewed Gamma	GHMM	0.8415	0.8708	0.18228
	GMM-HMM	**0.9837**	**0.9842**	**0.00685**

- Comparable classification under Gaussian emissions.
- GMM-HMM improves state recovery across non-Gaussian families.
- Lower emission KL accompanies substantially stronger classification.

Volatility-Defined Regimes: Fitted Densities Graph



Mean-Separated Regimes: Transition Recovery and Switching

	Transition RMSE		Average decoded switches	
Family	GHMM	GMM-HMM	GHMM	GMM-HMM
Gaussian	0.00088	0.00087	24.4	24.4
Student-t	0.01075	**0.00088**	73.6	24.7
Trimodal	0.00083	0.00086	24.1	24.2
Skewed Gamma	0.00167	**0.00088**	28.6	24.4

True average: 23.8 switches per test sequence.

- Both models closely recover transitions under Gaussian, trimodal, and skewed Gamma emissions.
- GHMM over-switches under Student-t emissions, despite its high classification accuracy.

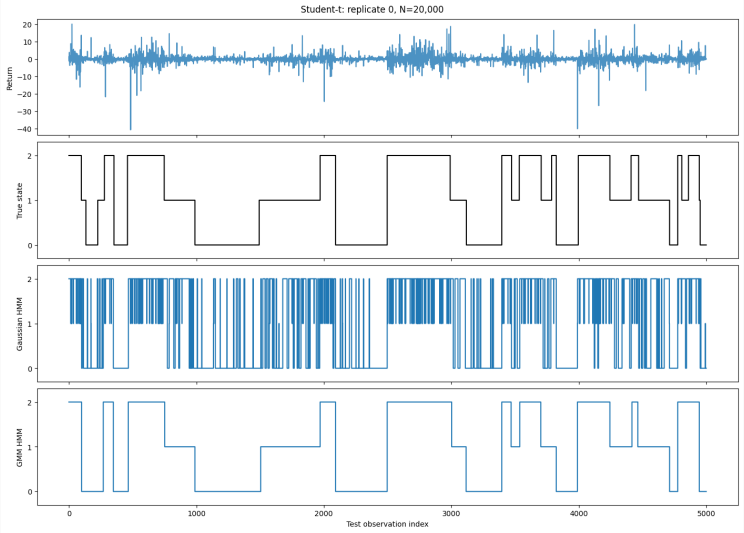
Volatility-Defined Regimes: Transition Recovery and Switching

	Transition RMSE		Average decoded switches	
Family	GHMM	GMM-HMM	GHMM	GMM-HMM
Gaussian	0.00071	0.00069	26.0	26.1
Student-t	0.40839	**0.00092**	486.4	23.7
Trimodal	0.25389	**0.00070**	542.1	28.4
Skewed Gamma	0.10875	**0.00081**	170.6	26.8

True average: 28.8 switches per test sequence.

- GHMM substantially over-switches and incorrectly estimates transitions under non-Gaussian emissions.
- GMM-HMM estimates transitions accurately and produces switch counts close to the truth.

Example of GHMM Over-Switching



Interpreting GHMM's performance degration on volatility regimes

- Mean-separated regimes provide location differences that help distinguish states despite emission misspecification.
- Volatility-defined regimes share the same mean. identifying them depends more on correctly representing distributional spread and tails.
- GMM-HMM remained robust to this while GHMM started to mix up regimes, causing rapid switching and inaccurate classification.
- Moreover, GHMM performed the worst on Student-t (heavy-tailed) data while GMM-HMM remained robust.

Examining the fitted state structure

Do the inferred states represent generating regimes, or different parts of their emission distributions?

Selected failures for Student-t: a state used for extreme observations

Two mean-separated Student-t replications

GHMM diagnostic	Replication 5	Replication 8
Test extreme observations captured	61.2%	56.5%

- A broad, low-persistence fitted state captures extreme observations.
- Another fitted state receives over 98% of training observations from *each* of two true regimes.
- Extremes satisfy $|x_t - \mu_{z_t}|/\sigma_{z_t} > 3$, using generating-state moments.

Consistent with states representing tail variation while true regimes merge.

Selected failures for Student-t: a state used for extreme observations

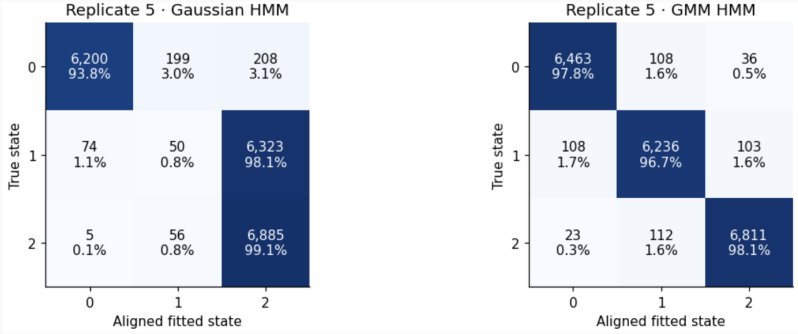


Figure 4: Confusion matrix of true versus aligned fitted state for GHMM and GMM-HMM on a replicate of Student-t emission data

Interpreting GHMM and GMM-HMM's performance on volatility and Student-t

- Gaussian emission was too restrictive for this case. GHMM wrongly merged majority of the observations in the tails of all states into one state trying to explain the variation of the model, while merging most of the observations in two true states into one regime.
- GMM-HMM was able to distinguish this through the use of its three components in each state and assigned most of tail observations to one component and the center observations to the other 2 components while preserving state structure. (i.e. mixture components can represent central and tail observations within a state)
- These cases illustrate how flexible emissions can capture within-regime variation while preserving the distinction between regimes, supporting more accurate state recovery.

Excessive Switching: Is it a problem?

- Usually, it signifies model overfitting to the noise
- Classification accuracy can hide this problem, brief incorrect excursions affect few observations but introduce multiple false transitions.
- In real-world financial applications, if regime changes trigger portfolio adjustments, false switches may increase turnover and transaction costs.
- This is highlighted by Shu et al. [8], who compared HMM-based strategy against JM-based (Jump Model) strategy, and hightlighted that JM achieved better risk-adjusted return metrics due to its lower tendency for excessive switching.

Conclusion on Synthetic Experiment

- GMM-HMM better approximates the non-Gaussian generating distributions, with comparable classification under Gaussian emissions.
- Both models recover persistent mean-separated states accurately, while GMM-HMM achieves substantially better recovery of non-Gaussian volatility regimes.
- GMM-HMM produces regimes that contains a lot less excessive switching and closer to the true switch counts compared to GHMM.
- High accuracy can coexist with excessive switching, inaccurate transitions, and poor emission-density fit.

Flexible emissions can improve regime recovery by representing complex variation within states, but improved synthetic results do not establish real-market usefulness.

S&P 500: Experimental Setup

- Compare Gaussian, Student-t, and GMM emissions with $M = 2$ or 3.
- Two hidden states, interpreted as relatively bullish and bearish conditions.
- Evaluate on 5,534 decision dates from 2004 to 2025.
- Start with ten years of training history, then expand the sample.
- Refit the scaler, PCA, and HMM every 63 decisions and use up to three EM updates between refits.

From Prices to Trading Positions

- Calculate daily simple returns, rolling downside deviation, and Sortino-style features.
- Use 5, 10, and 60-day rolling windows and EWMA half-lives.
- Standardize 28 features and apply PCA, retaining at least 95% of the variance.
- Forecast next-day bull probability with the HMM.
- Using 12-decision rolling median of raw bull threshold above 50%
- Hold the index when bullish, otherwise hold cash.

Feature construction: [8, 9]. PCA: [10].

Rolling Downside Deviation

Daily simple return and target-relative return:

$$r_t = \frac{P_t}{P_{t-1}} - 1, \qquad x_t = r_t - \tau, \qquad \tau = 0.$$

Clip nonnegative returns to zero: $z_t = \min(x_t, 0)$.

$$\bar{z}_{t,n} = \frac{1}{n}\sum_{j=0}^{n-1} z_{t-j}, \qquad D_{t,n} = \sqrt{\frac{1}{n}\sum_{j=0}^{n-1}(z_{t-j} - \bar{z}_{t,n})^2}.$$

- Use trailing windows $n \in \{5, 10, 60\}$ trading days.
- Calculate the population standard deviation of the clipped returns, including the zeros.

Sortino-Style Features

The trailing mean target-relative return is divided by downside deviation:

$$S_{t,n} = \frac{\frac{1}{n}\sum_{j=0}^{n-1} x_{t-j}}{\max(D_{t,n}, 10^{-6})}, \qquad n \in \{5, 10, 60\}.$$

- The denominator floor prevents division by zero.
- Also calculate EWMAs of returns, downside deviation, and Sortino-style features with 5, 10, and 60-day half-lives.
- Use only current and past observations, and drop incomplete warm-up rows before fitting.

Related feature construction: [8, 9].

Baseline Strategies

- **Buy-and-hold:** Stay fully invested in the S&P 500 throughout the evaluation period.
- **MA 50/200:** Hold the index when its trailing 50-day average price exceeds its trailing 200-day average; otherwise hold cash.

$$\overline{P}_{t,n} = \frac{1}{n}\sum_{j=0}^{n-1} P_{t-j}, \qquad w_t^{\mathrm{MA}} = \mathbf{1}\{\overline{P}_{t,50} > \overline{P}_{t,200}\}.$$

- Positions use information available through day t and earn the next day's return: $R_{t+1}^{\mathrm{strategy}} = w_t r_{t+1}$.
- Both baselines use the same evaluation dates as the HMM strategies. Cash earns zero; dividends and trading costs are excluded.

HMM Strategies: State Labels and Forecasts

- Compare Gaussian, Student-t, and GMM ($M = 2, 3$) emissions using the same features and trading rule.
- Label the state with the higher probability-weighted mean training return as **bull**; label the other state **bear**.
- Filter observations through day t to obtain $\alpha_{t,k} = P(z_t = k \mid x_{1:t})$.
- Use the transition matrix to forecast the next day's bull probability:

$$q_t = P(z_{t+1} = \text{bull} \mid x_{1:t}) = \sum_{k=1}^{2} \alpha_{t,k} A_{k,\text{bull}}, \qquad b_t = \mathbf{1}\{q_t \geq 0.5\}.$$

The trading forecast uses filtering, not a Viterbi path decoded using future observations.

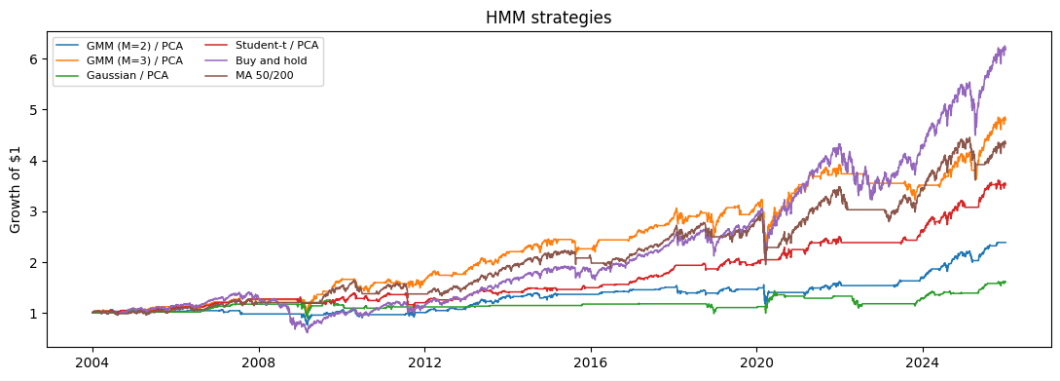
HMM Strategies: From Forecasts to Positions

- Smooth the binary forecasts b_t using a trailing 12-decision rolling median m_t.
- Hold the index when the median is above 0.5; hold cash when it is below 0.5. A tie preserves the previous position.

$$w_t = \begin{cases} 1, & m_t > 0.5, \\ 0, & m_t < 0.5, \\ w_{t-1}, & m_t = 0.5. \end{cases} \qquad R_{t+1}^{\mathrm{HMM}} = w_t r_{t+1}.$$

- Use available decisions during the initial 12-decision warm-up and initialize the previous position to cash.
- Long-only, fully invested or cash, and no short selling or leverage

Strategy Growth: 2004–2025



Buy-and-hold has the highest total return. Risk-adjusted performance differs.

Strategy Performance: 2004–2025

Strategy	Total return (%)	Sharpe	Max. drawdown (%)
Student-t	251.7	**0.640**	−21.4
GMM ($M = 3$)	379.6	0.544	−33.9
MA 50/200	332.9	0.494	−33.9
Buy-and-hold	**517.6**	0.439	−56.8
GMM ($M = 2$)	138.6	0.356	−29.2
Gaussian	60.6	0.243	−23.9

- Student-t and GMM ($M = 3$) have higher Sharpe ratios than both baselines.
- Returns exclude dividends, cash interest, and trading costs.

Do the Forecasts Separate Market Conditions?

Model	Bull: next-day return (%)	Bear: next-day return (%)
GMM ($M = 2$)	0.0343	0.0433
GMM ($M = 3$)	0.0581	0.0171
Gaussian	0.0441	0.0388
Student-t	0.0550	0.0274

- GMM ($M = 3$) and Student-t give clearer return separation.
- Their smoothed positions last 55.9 and 49.9 trading days on average, including both invested and cash periods.
- Persistent signals and higher returns do not establish regime accuracy: true market-state labels are unavailable.

Case Study: Signals During 2008

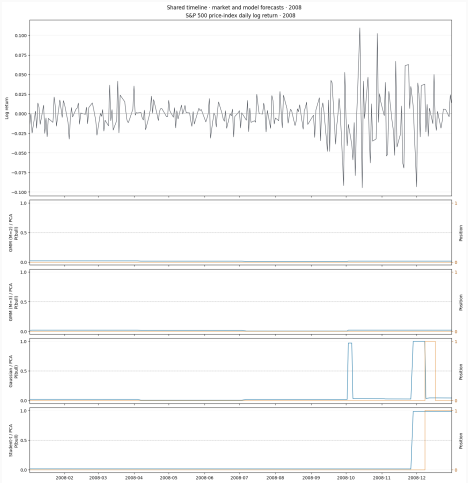
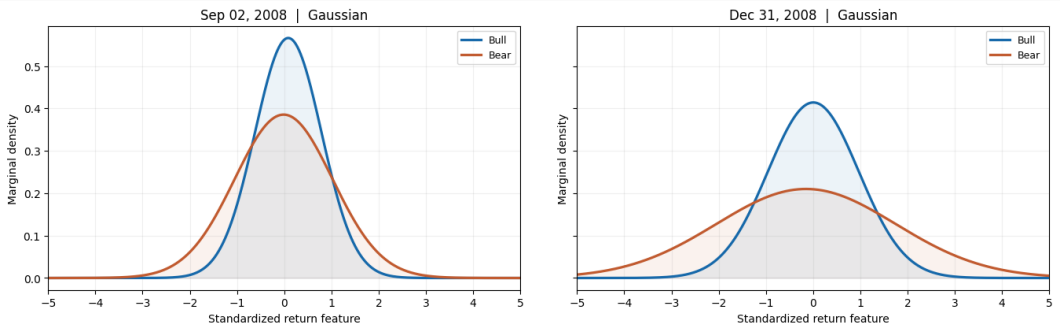


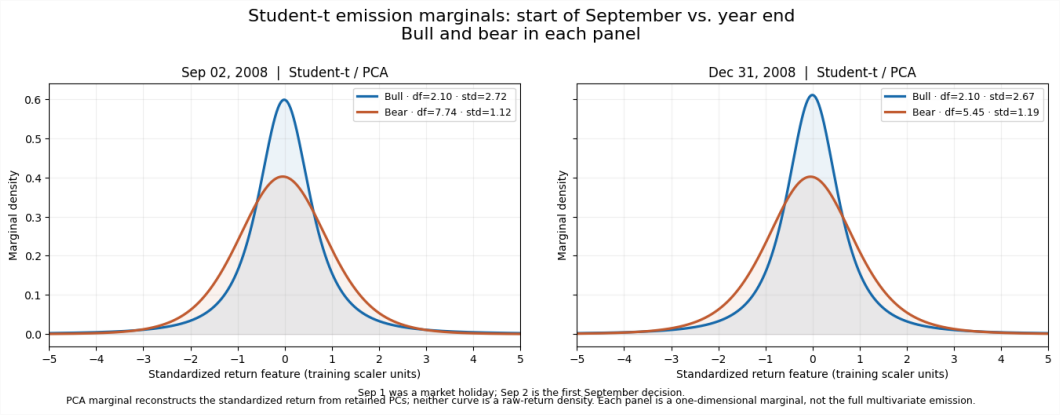
Figure 5: Blue and yellow lines indicate the b_t and 12-decision rolling median m_t ($m_t = 1$ means all-in S&P 500)

2008 Emission Densities: Gaussian



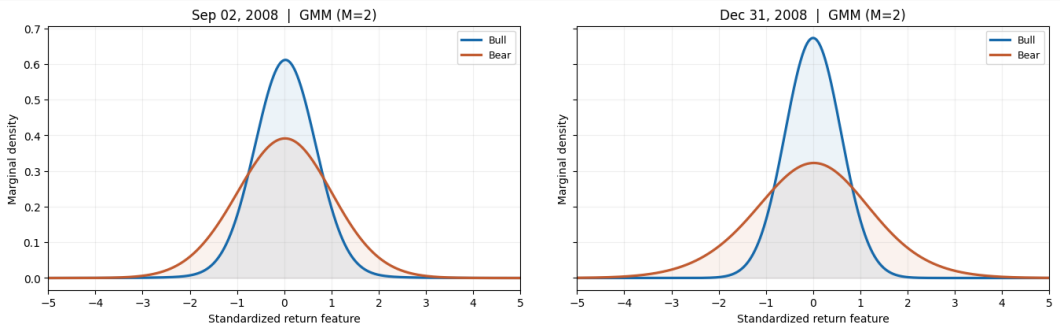
September 2 and December 31, 2008: bull- and bear-state marginal densities of the standardized return feature.

2008 Emission Densities: Student-t



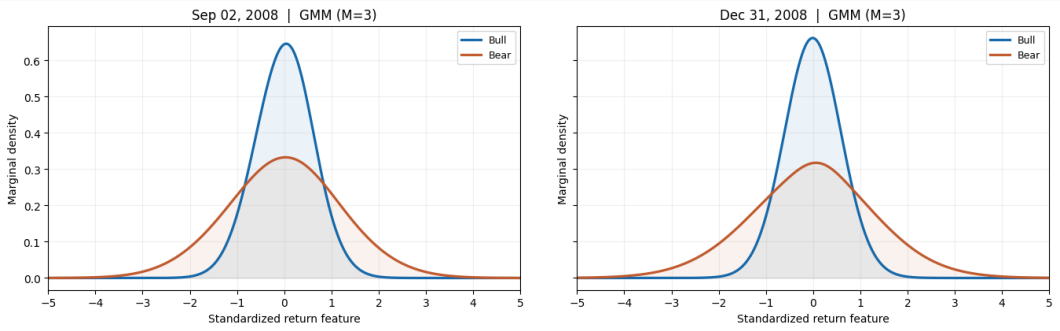
September 2 and December 31, 2008: bull- and bear-state marginal densities of the standardized return feature.

2008 Emission Densities: GMM ($M = 2$)



September 2 and December 31, 2008: bull- and bear-state marginal densities of the standardized return feature.

2008 Emission Densities: GMM ($M = 3$)

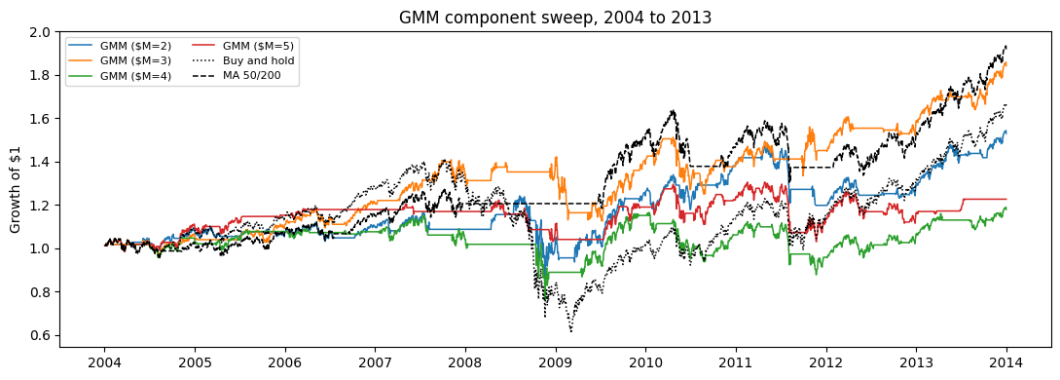


September 2 and December 31, 2008: bull- and bear-state marginal densities of the standardized return feature.

Changing the Number of GMM Components

- Compare $M = 2, 3, 4, 5$ components per state between 2004 and 2013
- Start with five years of training history and keep the same refit and signal-smoothing rules.
- All models still have two hidden states. More mixture components do not mean more market regimes.
- Examine both trading performance and the fitted component densities.

GMM Component Sweep: Strategy Growth



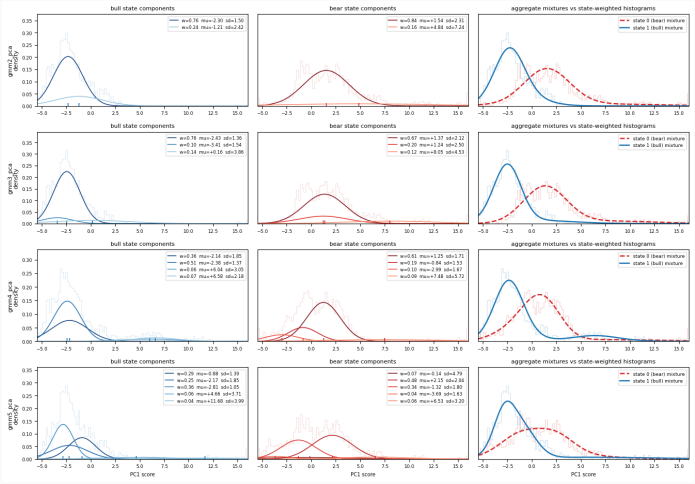
Three components give the strongest GMM strategy in this sweep, but MA 50/200 still performs better.

GMM Component Sweep: Results

Strategy	Return (%)	Sharpe	Switches/year	Days/position
GMM ($M = 2$)	53.0	0.277	5.41	45.76
GMM ($M = 3$)	84.2	0.507	5.41	45.76
GMM ($M = 4$)	17.9	0.110	4.61	53.55
GMM ($M = 5$)	22.7	0.191	4.91	50.34
MA 50/200	91.7	0.541	1.00	228.82
Buy-and-hold	65.3	0.246	0.00	–

- More components do not improve trading performance here.
- The four- and five-component models switch less often after smoothing, so their weaker results are not simply caused by more frequent trading.

Fitted GMM Densities Along PC1



Component Locations and Mixture Weights

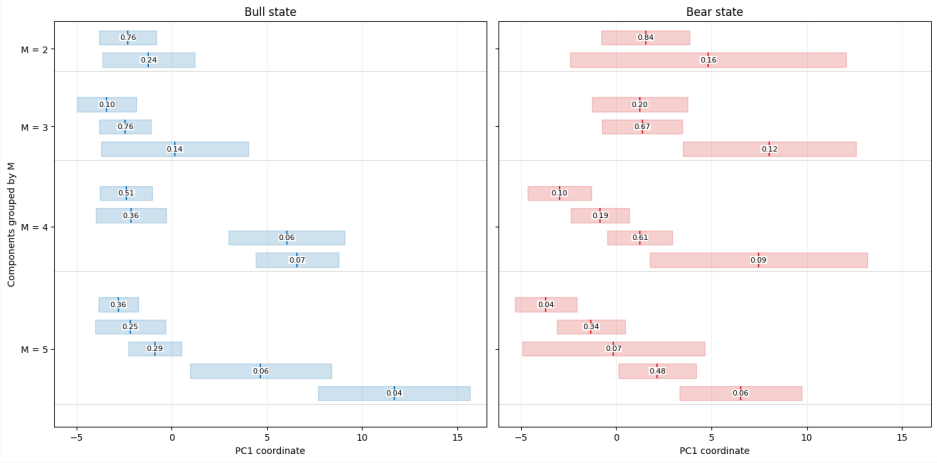


Figure 6: The weight of each component in GMM is denoted by the number in the interval, and intervals are drawn by plus minus one standard deviation of the mean of PC1

What Do the Components Represent?

- With two or three components, one component carries most of the bull-state weight. Smaller components describe variation around it or in the tails.
- With four or five components, more components overlap around the center, with small bull-state tail weights of roughly 0.04–0.07.
- These describe variation within the states, not separate market regimes.
- PC1 is not the market return. This projection alone cannot show whether the extra components fit useful structure or noise.
- The held-out-test is an experiment to explain the overfitting issue of GMM, but it's not included in this presentation.

S&P 500: Main Findings

- Student-t and the three-component GMM improve risk-adjusted strategy performance relative to Gaussian emissions in the full-period experiment.
- The three-component GMM gives the strongest GMM strategy in the shorter sweep. Adding more components does not help here.

Future Work: Normalizing Flow Emissions

- In the next few weeks, we'll focus on researching normalizing flows and its integration into HMM as an emission model.
- Normalizing flows are deep-learning generative models that transform a simple probability distribution into a more flexible distribution through a sequence of invertible transformations [11].
- Rather than restricting each state's emission density to a single Gaussian or a fixed finite Gaussian mixture, we will learn an invertible transformation of a base distribution for each state. The resulting densities can be evaluated through the change-of-variables formula and used within HMM inference.
- Given the limited observation amounts and the low signal-to-noise ratio of financial return data [12], we'll most likely need to investigate ways to prevent the NF-based emission model from overfitting to the noise.

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